Imagine you just had heart surgery.

You are exhausted.

Someone hands you 6 pages of discharge paperwork.

At 2am when something feels wrong, you have no idea what is normal and what means go to the ER. That moment is exactly what DischargeIQ is built for.

DischargeIQ

Multi-Agent Post-Discharge Patient Education System

NEED

The Discharge Failure

40-80%

of medical information forgotten immediately (vs recalled)

47%

of verbal discharge instructions correctly recalled by patients

$41.3B

annual US cost of preventable 30-day readmissions in the US

Patients leave the hospital without usable understanding of their own care plan.

- 95.6% of patients say they understood their discharge · only 59.6% can accurately describe their diagnosis 2 days later - a gap that current discharge processes do not detect
- Over 80% of Epic discharge docs exceed 6th-grade reading level · 36% of US adults read at or below that level · no system checks if patients understood
- Existing tools do not integrate diagnosis explanation, medication rationale, and escalation guidance into a single patient-specific output · that is the gap DischargeIQ fills

APPROACH

Why Multi-Agent Architecture?

We convert discharge data into structured understanding – one validated step at a time.

Single-Agent Prompt Chain

What it does

- All tasks handled in one prompt
- Lower latency, simpler to deploy

Limitations

- Couples extraction, reasoning, and explanation in one step
- Errors propagate without visibility or checkpoint
- Intermediate outputs cannot be validated or traced

VS

Multi-Agent Pipeline (Chosen)

Pipeline: Extraction → Diagnosis → Medication → Recovery → Escalation

Why this works

- Validation checkpoints between every step
- Structured JSON outputs verified at each handoff
- Failures are isolated and traceable to their source
- Structured validation across steps helps detect and prevent error propagation
- Each component can be tested and improved independently

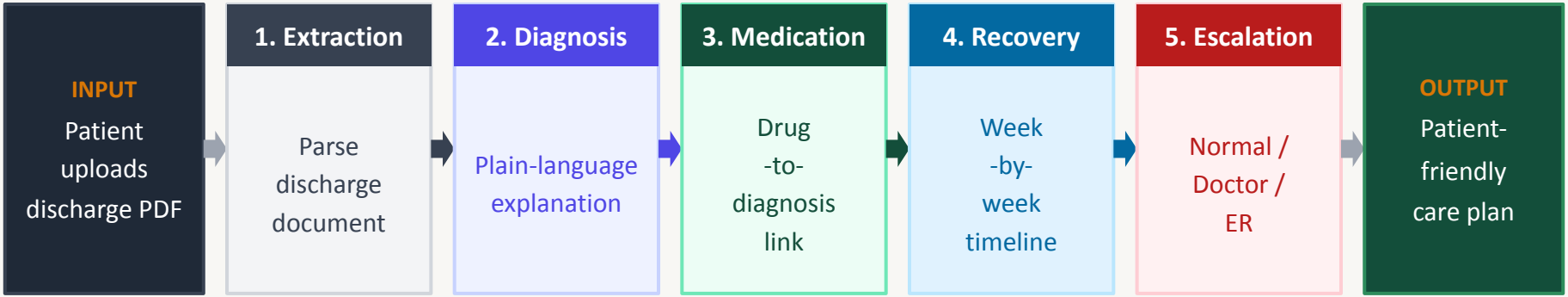
Design Principle: In healthcare systems, correctness > latency. The overhead of multiple calls is worth the traceability.

APPROACH

The 5-Agent Pipeline

We convert discharge data into structured patient understanding.

Each agent handles one task type · Structured JSON verified at every handoff · MVP targets 5 high-readmission diagnoses



Safety gate: Outputs outside the 5 validated templates surface a confidence warning rather than passing downstream. Null is better than wrong.

APPROACH

Key Innovation: What Each Agent Delivers

Component	What It Does	Example Output
Diagnosis	Plain-language explanation of what happened and why it matters	*"Your heart was not pumping enough to clear fluid from your lungs."*
Medication	Each drug linked to your specific condition — not just what it is, but WHY you are taking it	*"You take Furosemide because fluid buildup strains your heart."*
Recovery	Week-by-week timeline: what is expected vs. what is a warning sign	*"Week 1: mild fatigue is normal. Swelling in your legs is not."*
Escalation	Three-tier triage: normal / call your doctor / go to the ER now — clear thresholds, no guessing	*"Chest pain or sudden shortness of breath: call 911."*

DischargeIQ integrates these elements into a single patient-specific pipeline, which existing tools do not currently provide in combination.

BENEFITS

Value Delivered

The Proof Point

Flesch-Kincaid Reading Grade

Raw discharge doc — Grade 11.2

DischargeIQ output — Grade 5.1

6.1 grade level improvement

Measured on synthetic discharge scenarios across 4 conditions: COPD, CHF, pneumonia, hip replacement

💡 The Key Innovation: Diagnosis-to-Medication Connection

Understanding why a medication is prescribed improves adherence behavior. That connection is what every current tool skips

📋 Recovery Trajectory Reduces Unnecessary ER Visits

Normal vs. alarming symptom distinctions reduce panic calls for expected symptoms and accelerate action on genuinely dangerous ones

🏥 Direct CMS Readmission Penalty Impact for Hospitals

A measurable discharge quality intervention tied to the Hospital Readmissions Reduction Program - hospitals have a financial incentive, not just a moral one

MVP covers 5 high-readmission diagnoses. Clinical decision support is permanently excluded from scope - DischargeIQ explains, never advises. That distinction keeps it safe, deployable, and outside FDA oversight.

Understanding drives better outcomes.

COMPETITION

What Exists and What It Misses

Solution	DX Explanation	Medication Why?	Recovery Plan	Escalation Guide	Patient-Friendly
Epic AfterVisit Summary	✘	✘	✘	✘	✘
Verbal Discharge (Nurse)	✘	✘	✘	✘	✘
WebMD / Healthline	✘	✘	✘	✘	✘
DischargeIQ (Ours)	✔	✔	✔	✔	✔

DischargeIQ uniquely delivers all four components in a single patient-specific workflow.

RISKS

Risk Matrix

Risk	Probability	Impact	Why It Matters
LLM accuracy on edge case diagnoses	High	High	Outputs outside the 5 validated diagnosis templates could be inaccurate or misleading. A wrong medication explanation is a direct patient safety failure
Synthetic test data vs. real document messiness	High	Medium	Real discharge documents vary enormously across hospital systems. Synthetic test data may not expose failure modes that appear only on malformed or image-only PDFs
Scope creep into clinical decision support	Medium	High	The line between patient education and clinical advice is a regulatory boundary. Crossing it puts DischargeIQ in FDA Software as a Medical Device territory

MITIGATION

How We Address Each Risk

LLM accuracy on edge cases → Hard scope limit to 5 clinician-reviewed diagnosis templates with guideline citations. Anything outside those 5 surfaces a partial confidence warning rather than a hallucinated answer. Null is better than wrong - always.

Synthetic vs. real data gap → Evaluation framework runs 20 test cases through an LLM-as-Judge rubric with a Clinical Accuracy target and a hard Safety pass/fail gate. Any Safety failure is a P0 fixed before the demo, not documented as a known issue.

Scope creep into clinical advice → Clinical decision support is permanently excluded from scope, not deferred to a future sprint. Every agent prompt is written with that constraint built in. This keeps DischargeIQ in patient education territory and outside FDA device regulation.

CONCLUSION

Why This Matters

DischargeIQ fills the critical gap between a hospital's discharge paperwork and a patient's ability to follow it safely at home. Every current tool leaves patients with information they cannot use. We designed a system that transforms discharge information into actionable patient understanding.

Next Steps

01 Complete MVP

Full 5-agent pipeline covering all target diagnoses. LLM-as-Judge evaluation. Flesch-Kincaid gate on every output.

02 Pilot Test

Clinical setting pilot with real discharge documents. Measure comprehension improvement vs. baseline AfterVisit summaries.

03 Feed LOF Ecosystem

Structured JSON output feeds into TheraCareAI as patient baseline comprehension data, enabling more effective downstream adherence tracking.

Turning the discharge knowledge gap into a solvable engineering problem, one patient at a time.